

thus simulate light touch. This corresponds to our sense of the fine hairs on our skin being gently stroked," said Philip Mittendorf, a scientist who is developing the artificial skin.

There are also six temperature sensors and an accelerometer. This allows the machine to register the movement of individual limbs, for example, of its arms, and thus to learn what body parts it has just moved. Plate for plate, the boards are placed together forming a honeycomb-like, planar structure to be worn by the robot. For the machine to have detection ability, the signals from the sensors must be processed by a central computer. This enables each sensory module to not only pass its own information but to also serve as a data hub for different sensory elements. This happens automatically, ensuring that signals can go in alternative ways if a connection should fail.

Guide vests—robotic navigation aids for the visually impaired

A group of USC Viterbi School of Engineering researchers are creating a high-tech aid for blind and partially blind people who have long depended on white canes as their only tool for getting around.

The system uses cameras worn on the head connected to PCs that use simultaneous localisation and mapping



A user wearing the guide vest fitted with binocular camera

(SLAM) software to build maps of the environment and identify a safe path through obstacles. This route information is conveyed to the user through a guide vest that includes four micro motors located on an individual's shoulder and waist that vibrate like cell phones.

SLAM software analyses data from stereo camera views to create 3D renderings of the scene and then map a path through it. For example, a vibration on the left shoulder indicates a higher object to the left, such as a low-hanging branch, and the individual can in turn use that information to take a new path.

The current head-mounted camera is bulky, and the team is now working on a micro-camera system that could be attached to glasses.

A lens-free, pinhead-size camera

Scientists at the Cornell University have developed a Brownie-like camera. The microscopic device fits on the head of a pin and contains no lenses or moving parts.

The camera was invented in the lab of Alyosha Molnar, assistant professor of electrical and computer engineering, and developed by a group led by Patrick Gill, a postdoctoral associate. The camera's working prototype is one-hundredth of a millimetre thick, and one-half millimetre on each side. The camera resolves images about 20-pixels across. It is just a flat piece of doped silicon, which looks something like a tiny CD, with no parts that require off-chip manufacturing.

The scientists call their camera a planar fourier capture array (PFCA) because it uses the principles of the fourier transform, which is a mathematical tool that allows multiple ways of capturing the same information. Each pixel in the PFCA reports one

component of the fourier transform of the image being detected by being sensitive to a unique blend of incident angles. While Fourier components themselves are sometimes directly useful, a bit of computation can also transform the components into an image.

The camera could be used as a component in any cheap electronics system—in devices that, for example, detect the angle of the sun or a micro-robot that requires a simple visual system to navigate.

World's fastest mainboard with integrated graphics

MSI has come up with the world's fastest mainboard with integrated graphics—A75MA-G55.

The mainboard incorporates MSI's OC Genie II technology, which allows single-click over-clocking without the need of any technical knowledge. OC Genie II is available on all MSI A75-Series mainboards. MSI claims that the mainboard has broken records on 3D Mark 06, which has been



MSI A75MA-G55

made possible only by versatile Click BIOS UEFI interface and improvements in mainboard design.

MSI's Military Class II components have been certified by the United States Department of Defense 810G standard. Only Military Class II components have the lifetime and stability required by today's most demanding situations of extreme over-clocking but at the same time provide great levels of power saving.

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